

ALGEBRA

1. Let $GL(n, q)$ be the group of all $n \times n$ invertible matrices over the finite field $\mathbb{F}_q$, where $q = p^m$, p is a prime and some positive integer m . Then
 - (i) $O(GL(n, q)) = q^{\frac{n(n-1)}{2}}(q^n - 1)(q^{n-1} - 1) \cdots (q - 1)$.
 - (ii) Order of Sylow p -subgroup of $GL(n, q)$ is $q^{\frac{n(n-1)}{2}}$.
 - (iii) $O(SL(n, q)) = \frac{q^{\frac{n(n-1)}{2}}(q^n - 1)(q^{n-1} - 1) \cdots (q - 1)}{(q - 1)}$.

2. The center $Z(GL(n, q)) = \{kI_n \mid k \in \mathbb{F}_q^*\}$, where I_n is the identity matrix.

3. The centralizer (normalizer)

$$N(SL(n, q)) = \{kI_n \mid k \in \mathbb{F}_q^* \text{ and } k^n = 1\}.$$

4. $Z(SL(n, q)) = \{kI_n \mid k \in \mathbb{F}_q^* \text{ and } k^n = 1\}$.

5. $|Z(SL(n, q))| = \gcd(n, q - 1)$.

6. Let G be a finite group and $a \in G$. Then $O(cl(a)) = \frac{O(G)}{O(N(a))}$.

7. $\frac{G}{Z(G)} \cong I(G)$.

8. In S_n , the number of distinct cycles of length r is $\frac{n!}{r(n-r)!}$ ($r \leq n$).

9. Converse of Lagrange's theorem holds in finite cyclic groups and prime power order.

10. $Z(G) = \bigcap_{a \in G} N(a)$.

11. The number of group homomorphism from $\mathbb{Z}_m$ to $\mathbb{Z}_n$ is $\gcd(m, n)$.

12. Let G be an infinite cyclic group. Then $|Aut(G)| = 2$.

13. Let G be a finite cyclic group of order n . Then $|Aut(G)| = \varphi(n)$.

14. $Aut(S_n) \cong S_n$, $Z(S_n) = \{I\}$ for $n \geq 3$ and $n \neq 6$.

15. $Aut(\underbrace{\mathbb{Z} \oplus \mathbb{Z} \oplus \cdots \oplus \mathbb{Z}}_{n \text{ times}}) \cong GL(n, \mathbb{Z})$.

16. $Aut(\underbrace{\mathbb{Z}_{p^m} \oplus \mathbb{Z}_{p^m} \oplus \cdots \oplus \mathbb{Z}_{p^m}}_{n \text{ times}}) \cong GL(n, \mathbb{Z}_{p^m}).$
17. $Aut(\mathbb{Z}_2 \oplus \mathbb{Z}_4) \cong D_8.$
18. If G has order $n > 1$, then $|Aut(G)| \leq \prod_{i=0}^k (n - 2^i)$ where $k = \lceil \log_2(n - 1) \rceil$.
19. Let p be a prime number, and let G be a finite abelian group in which every element has order p . Then $Aut(G) \cong GL(n, \mathbb{Z}_p)$, where n is the dimension of G over $\mathbb{Z}_p$.
20. If G is a group of order n and F is any field, then $GL(n, F)$ contains a subgroup isomorphic to G .
21. Let $G = G_1 \times G_2 \times \cdots \times G_n$, where the G_i are abelian groups. Then $Aut(G)$ is isomorphic with the group of all invertible $n \times n$ matrices whose (i, j) entries belong to $Hom(G_i, G_j)$, the usual matrix product being the group operation.
22. If $O(G) = p^2q^2$ and $q \nmid p^2 - 1$, $p \nmid q^2 - 1$, then G is abelian.
23. G is a finite group of order p^2q where p and q are distinct primes such that $p^2 \not\equiv 1 \pmod{q}$ and $q \not\equiv 1 \pmod{p}$. Then G is an abelian group. If p divides $q - 1$, then any group of order p^2q is abelian.
24. If p does not divide $|Aut(G)|$, then any group of order pq^2 is abelian.
25. If G is a non-abelian finite group, then $|Z(G)| \leq \frac{1}{4}|G|$.
26. If H and K are subgroups of a finite group G satisfying $(|G : H|, |G : K|) = 1$, then $G = HK$.
27. If G is a simple group of order 60, then G is isomorphic to A_5 .
28. Let G be a group of order pqr , where $p > q > r$ are primes. If $p - 1$ is not divisible by q or r and $q - 1$ is not divisible by r , then G must be abelian (hence cyclic).
29. Abelian groups have exactly one Sylow p -subgroup for each p .
30. The class equation of G is

$$O(G) = O(Z(G)) + \sum_{a \notin Z(G)} \frac{O(G)}{O(N(a))}.$$

31. Let G be a non-abelian group of order p^3 . The number of conjugate classes of G is $p^2 + p - 1$.
32. Let G be a finite group of order n and p be a prime number such that $p > \frac{n}{p}$. Then any subgroup of order p in G is normal in G .
33. Let G be a finite group of order n and p be a prime number such that p^2 does not divide n . Then any subgroup of order p in G is normal in G .
34. The number of non-isomorphic abelian groups of order p^n , (p a prime) is $p(n)$ (partition of n).
35. The number of groups of order n is at most $n^{n \log_2 n}$.
36. A group of order n can be generated by a set of at most $\log_2 n$ elements.
37. Let $n > 1$. The content of n is defined to be the product of all distinct primes dividing n and is denoted by $c(n)$. $a \in \mathbb{Z}_n$ is nilpotent **iff** $c(n)$ divides a .
38. For any integers a and n with $0 \leq a < n$, then a is idempotent in $\mathbb{Z}_n$ **iff** $a(a-1)$ is a multiple of n .
39. Let $f(x) = a_0 + a_1x + \cdots + a_{q-2}x^{q-2} \in \mathbb{F}_q[x]$. Then the number of non-zero solutions of the polynomial in $\mathbb{F}_q$ is equal to $q-1-r$, $r = \text{rank}(A)$

$$\text{where } A = \begin{pmatrix} a_0 & a_1 & a_2 & \cdots & a_{q-2} \\ a_1 & a_2 & a_3 & \cdots & a_0 \\ a_2 & a_3 & a_4 & \cdots & a_1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{q-2} & a_0 & a_1 & \cdots & a_{q-1} \end{pmatrix}_{(q-1) \times (q-1)}$$

40. Any ring of prime order is commutative.
41. A ring of order p^2 (p a prime) may not be commutative.
42. Smallest non commutative ring is of order 4.
43. $Ch(R \times S) = l.c.m(ChR, ChS)$.
44. $\mathbb{Z}[\sqrt{-19}]$ is Principle Ideal Domain but not Euclidean Domain.
45. A field $(\mathbb{F}, +, \cdot)$ is always a Unique Factorization Domain.

46. $\mathbb{Z}[\sqrt{-5}]$ is an integral domain which is not a Unique Factorization Domain.
47. Let $\mathbb{F}$ be a field. Then $\mathbb{F}[x, y]$ is an example of a Unique Factorization Domain D which is not a Principle Ideal Domain.
48. $\mathbb{Z}_n$ has no non-zero nilpotent iff n is square free.
49. If R has unity 1, then $1 + I$ is unity of $\frac{R}{I}$.
50. Field of reals can be imbedded into the field of complex numbers.
51. If $x \in [0, 1]$, then $M = \{f \in C[0, 1] \mid f(x) = 0\}$ is a maximal ideal of $C[0, 1]$.
52. If F is a field, then the kernel of any evaluation map $F[x] \longrightarrow F$ is a maximal ideal.
53. A Euclidean domain possesses unity.
54. Every field is a Euclidean domain.
55. For any commutative ring R with unity, $\frac{R[x]}{\langle x \rangle} \cong R$.
56. An integral domain R with unity is a field **iff** $R[x]$ is a Principle Ideal Domain.
57. The order of Sylow p -subgroup of S_n is p^k where $k = \left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + \cdots$.
58. Let $\sigma = (1, 2, 3, \dots, n) \in S_n$. The number of elements in the conjugate class of σ is $(n-1)!$.
59. If G is a group and A is an abelian group, then there is a bijection between $\text{Hom}(G, A)$ and $\text{Hom}\left(\frac{G}{[G, G]}, A\right)$ where $G' = [G, G] = \langle aba^{-1}b^{-1} \rangle$.
60. $S'_n = A_n$ for $n \geq 1$, $A'_n = A_n$ for $n \geq 5$ and $A'_4 = K_4$.
61. $(GL(n, \mathbb{C}))' = SL(n, \mathbb{C}) = (SL(n, \mathbb{C}))'$.
62. For $n \geq 3$, D'_{2n} is $\langle s^2 \rangle$ or $\langle s \rangle$ according as n is even or odd.
63. If a normal subgroup N of A_n ($n \geq 3$) contains a 3-cycle, then $N = A_n$.

64. $\frac{A_4}{V_4}$ is cyclic group of order 3.

65. $\frac{S_4}{V_4} \cong S_3$.

66. $\frac{\mathbb{Q}}{\mathbb{Z}} \cong \mu_\infty = \{\text{all roots of 1 in } \mathbb{C}\}$.

67. $\frac{\mathbb{C}}{\mathbb{R}} \cong \mathbb{R}, \frac{\mathbb{C}^*}{\mathbb{R}^*} \cong S^1$.

68. $\frac{\mathbb{C}^*}{S^1} \cong \mathbb{R}_+^*$.

69. $\frac{GL(n, \mathbb{C})}{SL(n, \mathbb{C})} \cong \mathbb{C}^*$.

70. $\frac{GL(n, \mathbb{Z})}{SL(n, \mathbb{Z})} \cong \{-1, 1\}$.

71. $\frac{U(n, \mathbb{C})}{SU(n, \mathbb{C})} \cong S^1$.

72. $\frac{O(n, \mathbb{C})}{SO(n, \mathbb{C})} \cong \{-1, 1\}$.

73. Let G be an abelian group of order p^α .

(i) $|Aut(G)| = (p^\alpha - 1)(p^\alpha - p) \cdots (p^\alpha - p^{\alpha-1})$.

(ii) The number of subgroups of G of order p^β , $1 \leq \beta \leq \alpha$, is

$$\frac{(p^\alpha - 1)(p^{\alpha-1} - 1) \cdots (p^{\alpha-\beta+1} - 1)}{(p^\beta - 1)(p^{\beta-1} - 1) \cdots (p - 1)}.$$

74. The number of ring homomorphism from $\mathbb{Z}_n$ to $\mathbb{Z}_m$ is $2^{w(m)-w(\frac{m}{(n,m)})}$, where $w(n)$ denotes the number of prime divisors of n .

75. $\frac{D_{2n}}{\langle r^k \rangle} \cong D_{2k}$.

76. In D_{2n} ,

$$O(Z(D_{2n})) = \begin{cases} 2 & \text{if } n \text{ is even} \\ 1 & \text{if } n \text{ is odd.} \end{cases}$$

77. In D_{2n} ,

$$\text{Number of elements of order } 2 = \begin{cases} n+1 & \text{if } n \text{ is even} \\ n & \text{if } n \text{ is odd.} \end{cases}$$

78. Number of cyclic subgroup of order $n = \frac{\text{No. of elements of order } n}{\varphi(n)}$.

79. If G is a finite abelian group having a unique subgroup of order p for every prime divisor p of $O(G)$, then G is cyclic.

80. The group of n^{th} roots of unity is cyclic.

81. A generator for this group is called a primitive n -th root of unity.

82. Let ξ be a primitive n^{th} root of unity, then $[\mathbb{Q}(\xi); \mathbb{Q}] = \varphi(n)$.

83. The Galois group of the general polynomial of degree n is S_n .

84. $\text{Aut}(\mathbb{Z}_n) \cong U(n)$.

85. The number of permutation in S_n ($n \geq 4$) which commute with $\sigma = (12)(34)$ is $8(n-4)!$.

86. If $\sigma = (12)(34)$ in S_n , then $|cl(a)| = \frac{n(n-1)(n-2)(n-3)}{8}$.

87. Let $\sigma \in S_n$ and let $m_1, m_2, \dots, m_s$ be distinct integers which appear in the cycle type of σ (including 1-cycles). For each $i \in \{1, 2, \dots, s\}$ assume σ has k_i cycles of length m_i (so that $\sum_{i=1}^s k_i m_i = n$). The number of conjugates of σ is

$$\frac{n!}{(k_1! m_1^{k_1})(k_2! m_2^{k_2}) \cdots (k_s! m_s^{k_s})}.$$

88. Let σ be the m -cycle $(1\ 2 \cdots m)$. Then σ^i is also an m -cycle **iff** $(i, m) = 1$.

89. If $G = \langle x \rangle$ be a cyclic group of order n . Then a subgroup H is normal **iff** $H = \langle x^p \rangle$ for some prime $p|n$.

90. In D_{2n} , if $k|n$, $\langle r^k \rangle$ is normal subgroup in D_{2n} .

91. If $|G : H| = m$, $|G : K| = n$. Then $\text{lcm}(m, n) \leq |G : H \cap K| \leq mn$.

92. If $H \leq K \leq G$, then $|G : H| = |G : K||K : H|$.
93. If N is a normal subgroup of the finite group G and $\left(|N|, \frac{|G|}{|N|}\right) = 1$, then N is the unique subgroup of G of order $|N|$.
94. If G is a simple group of odd order, then $G \cong \mathbb{Z}_p$ for some prime p .
95. The finite group G is solvable **iff** for every divisor n of $|G|$ such that $\left(n, \frac{|G|}{n}\right) = 1$, G has a subgroup of order n .
96. If p is prime, then $S_p = \langle \sigma, \tau \rangle$, σ is any transposition and τ is any p cycle.
97. If σ is an m -cycle in S_n . Then $|N(\sigma)| = m(n - m)!$.
98. Let p and q be prime numbers with $p < q$, and let d be the smallest number such that $p^d \equiv 1 \pmod{q}$. Then, any group of order $p^k q$ where $1 \leq k < d$ contains a normal subgroup of order q .
99. If $|G| = p^d q$, then either G contains a normal subgroup of order q or a normal subgroup of order p^d .
100. Let G be a group with a subgroup H of index n . If $n! < |G|$, then the group G is not simple.
101. $Z(\mathbb{Q}_8) = \langle -1 \rangle$, $\text{Inn}(\mathbb{Q}_8) \cong K_4$.
102. $Z(D_8) = \langle r^2 \rangle$, $\text{Inn}(D_8) \cong K_4$.
103. $U(2^n) \cong \mathbb{Z}_2 \oplus \mathbb{Z}_{2^{n-2}}$, $n \geq 3$.
104. Let p be a prime and let V be an abelian group with the property that $pv = 0$ for all $v \in V$ and $|V| = p^n$. Then $\text{Aut}(V) \cong GL_n(\mathbb{F}_p)$.
105. For all $n \neq 6$, we have $\text{Aut}(S_n) = \text{Inn}(S_n) \cong S_n$.
For $n = 6$, we have $|\text{Aut}(S_6) : \text{Inn}(S_6)| = 2$.
106. $\text{Aut}(D_8) \cong D_8$, $\text{Aut}(\mathbb{Q}_8) \cong S_4$.
107. Sylow p -subgroup of D_{2n} is cyclic and normal for every odd prime p .
108. $\frac{SL_2(\mathbb{F}_3)}{Z(SL_2(\mathbb{F}_3))} \cong A_4$.

109. If N is a normal subgroup of G , then $n_p(\frac{G}{N}) \leq n_p(G)$.
110. $SL_2(\mathbb{F}_4) \cong A_5$.
111. $n_p(A \times B) = n_p(A)n_p(B)$, where A, B are finite groups.
112. If x is a nilpotent, then $1 + x$ is unit.
113. Sum of unit and nilpotent is unit.
114. $m\mathbb{Z} + n\mathbb{Z} = d\mathbb{Z}$, $d = \gcd(m, n)$.
115. $m\mathbb{Z} \cap n\mathbb{Z} = l\mathbb{Z}$, $l = \text{lcm}(m, n)$.
116. If $q \in \mathbb{Z}$ be a prime with $q \equiv 3(\text{mod } 4)$. Then $\frac{\mathbb{Z}[i]}{\langle q \rangle}$ is a field with q^2 elements.
117. Let π be an irreducible element in $\mathbb{Z}[i]$. Then $\left| \frac{\mathbb{Z}[i]}{\langle \pi^n \rangle} \right| = \left| \frac{\mathbb{Z}[i]}{\langle \pi \rangle} \right|^n$.
118. If F is a field, then the polynomial ring $F[x]$ is a Euclidean domain.
119. $\mathbb{R}[x^2, x^3]$ is not Euclidean domain.
120. If R is Unique Factorization Domain, then a polynomial ring in an arbitrary number of variables with coefficients in R is also a Unique Factorization Domain.
121. Let $p(x) = a_0 + a_1x + \cdots + a_nx^n \in \mathbb{Z}[x]$. If $\frac{r}{s} \in \mathbb{Q}$, $(r, s) = 1$ and $p(\frac{r}{s}) = 0$, then $r|a_0$ and $s|a_n$.
122. If $p(x)$ is a monic polynomial with integer coefficients and $p(d) \neq 0$ for all integers d dividing the constant term of $p(x)$, then $p(x)$ has no roots in $\mathbb{Q}$.
123. A polynomial of degree two or three over a field F is irreducible **iff** it has no root in F .
124. An irreducible polynomial over $\mathbb{F}_q$ of degree n remains irreducible over $\mathbb{F}_{q^k}$ iff $(k, n) = 1$.
125. If $\sigma \in S_n$ is the product of disjoint cycles of lengths $n_1, n_2, \dots, n_r$ with $n_1 \leq n_2 \leq \cdots \leq n_r$ (including its 1-cycles), then the integers $n_1, n_2, \dots, n_r$ are called the **cycle type** of σ .

126. Two elements of S_n are conjugate in S_n **iff** they have the same cycle type.
127. In a integral domain primes implies irreducibility.
128. Field have no primes and irreducible element.
129. Let $\phi : F \rightarrow F$ be a homomorphism of fields. Then ϕ is either identically 0 or is injective, so that the image of ϕ is either 0 or isomorphic to F .
130. If $f(x) \in F[x]$ has n distinct roots in its splitting field E , then $Gal(E/F)$ is isomorphic to a subgroup of the symmetric group S_n , and so its order is a divisor of $n!$.
131. $Gal(\mathbb{Z}_{p^n}|\mathbb{Z}_p) \cong \mathbb{Z}_n$.
132. If F is a field and $E = F(\alpha)$, where α is a primitive n^{th} root of unity, then $Gal(E/F)$ is isomorphic to a subgroup of $U(n)$ and hence $Gal(E/F)$ is an abelian group.
133. Let F be finite field of order p^n . Then the $Aut(F)$ is cyclic group of order n .
134. A polynomial $f(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_0$ is **reduced** if $c_{n-1} = 0$. If $f(x)$ is a monic polynomial of degree n and if $c_{n-1} \neq 0$ in F , then its associated reduced polynomial is $\widetilde{f(x)} = f(x - \frac{c_{n-1}}{n})$.
135. A polynomial $f(x)$ and its associated reduced polynomial $\widetilde{f(x)}$ have same discriminant.
136. The discriminant of a reduced cubic $\widetilde{f(x)} = x^3 + qx + r$ is $D = -4q^3 - 27r^2$.
137. If $f(x) = x^3 + ax^2 + bx + c$, then its associated reduced polynomial is $x^3 + qx + r$, where $q = b - \frac{a^2}{3}$ and $r = \frac{2a^3}{27} - \frac{ab}{3} + c$.
138. The discriminant of $f(x)$ is $D = a^2b^2 - 4b^3 - 4a^3c - 27c^2 + 18abc$.
139. Let $f(x) \in \mathbb{Q}[x]$ be an irreducible cubic with Galois group G and discriminant D .
 - (i) $f(x)$ has exactly one real root **iff** $D < 0$, in which case $G \cong S_3$.
 - (ii) $f(x)$ has three real roots **iff** $D > 0$. In this case, either $\sqrt{D} \in \mathbb{Q}$ and $G \cong \mathbb{Z}_3$ or $\sqrt{D} \notin \mathbb{Q}$ and $G \cong S_3$.

140. Let n be a positive integer. The sequence of n consecutive composite numbers is $(n+1)! + 2, (n+1)! + 3, \dots, (n+1)! + (n+1)$.
141. The area of the polygon whose vertices are the n^{th} roots of unity in the complex plane, when $n \geq 3$ is $\frac{n}{2} \sin \frac{2\pi}{n}$.
142. Let p be a prime and let n be a positive integer, if $m \geq 0$ and if $\sum$ is the sum of the p -adic digits of m , then

$$n^m = n^{\sum} \pmod{p}.$$

143. $n^3 = n \pmod{6}$ for all $n \in \mathbb{Z}$.
144. $\gcd(a, b)$ is the smallest positive integer of the form $as + bt$.
145. Let n and a be positive integers and let $d = \gcd(a, n)$.
Then $ax \equiv 1 \pmod{n}$ has a solution **iff** $d = 1$.
146. Let G be a group and let a be an element of G of order n . For each integer k between 1 and n , then $|a^k| = |a^{n-k}|$.
147. $U(2^n)$, ($n \geq 3$) is not cyclic.
148. The number of subgroup of the dihedral group D_n ($n \geq 3$) is $d(n) + \sigma(n)$.
Where $d(n)$ is the number of positive divisors of n and $\sigma(n)$ is the sum of the positive divisors of n .
149. Every subgroup of D_n of odd order is cyclic.
150. The group of rational numbers under addition, has no proper subgroup of finite index.
151. Suppose $(t, s) = 1$. Then $U(st) \simeq U(s) \oplus U(t)$.
152. $U(2^n) \simeq \mathbb{Z}_2 \oplus \mathbb{Z}_{2^{n-2}}$, $n \geq 3$.
153. $U(p^n) \simeq \mathbb{Z}_{p^n - p^{n-1}}$, p odd prime.
154. The centre of a group is characteristic.
155. If $\frac{G}{Z(G)}$ is cyclic, then G is abelian.
156. If G is non-abelian, then $\text{Aut}(G)$ is not cyclic.
157. $\frac{\mathbb{Z} \oplus \mathbb{Z}}{\langle (a, 0) \rangle \times \langle (0, b) \rangle} \simeq \mathbb{Z}_a \oplus \mathbb{Z}_b$.

158. If A is finite, then

$$A \oplus B \simeq A \oplus C \text{ iff } B \oplus C.$$

159. $\frac{D_4}{Z(D_4)} \simeq \mathbb{Z}_2 \oplus \mathbb{Z}_2.$

160. Let G be the group $\frac{\mathbb{Q}}{\mathbb{Z}}$. Then for every positive integer t , G has a unique cyclic subgroup of G .

161. If $1 + k$ is idempotent in $\mathbb{Z}_n$, then $n - k$ is an idempotent in $\mathbb{Z}_n$.

162. If I is an maximal ideal of a commutative ring R with unity, then $I[x]$ need not be maximal ideal in $R[x]$.

163. If I is a prime ideal of a commutative ring R with unity, then $I[x]$ is a prime ideal of $R[x]$.

164. Let $f(x)$ and $g(x)$ be irreducible polynomial over a field F . If $f(x)$ and $g(x)$ are not associates, then

$$\frac{F[x]}{\langle f(x)g(x) \rangle} \simeq \frac{F[x]}{\langle f(x) \rangle} \oplus \frac{F[x]}{\langle g(x) \rangle}.$$

165. In $\mathbb{Z}_n$, there are 2^d idempotent elements, where d is the number of distinct prime divisors of n .

166. Sum of positive integers less than or equal to n and relatively prime to n is $\frac{n\phi(n)}{2}$.

167. If $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$, then the sum of all positive divisors of n is

$$\sigma(n) = \left(\frac{p_1^{\alpha_1+1} - 1}{p_1 - 1} \right) \left(\frac{p_2^{\alpha_2+1} - 1}{p_2 - 1} \right) \cdots \left(\frac{p_k^{\alpha_k+1} - 1}{p_k - 1} \right).$$

168. If $2 \neq d|n$, then the number of elements of order d in D_{2n} is $\phi(d)$.

169. Number of elements of order d in $\mathbb{Z}_n \times \mathbb{Z}_m$ is $\sum_{d_1|n, d_2|m} \phi(d_1)\phi(d_2)$, where $d = lcm(d_1, d_2)$.

170. If $k|n$ and $U_k(n) = \{x \in U(n) \mid x \equiv 1(mod k)\}$, then $\frac{U(n)}{U_k(n)} \simeq U(k)$.

171. $\frac{\mathbb{Z} \times \mathbb{Z}}{\langle (a, b) \rangle}$ is not cyclic if $\gcd(a, b) \neq 1$.
172. Number of elements of order n in $\frac{\mathbb{Q}}{\mathbb{Z}}$ is $\phi(n)$.
173. Number of elements of order d in S_n is $\frac{n!}{1^{\alpha_1} 2^{\alpha_2} \dots k^{\alpha_k} \alpha_1! \alpha_2! \dots \alpha_k!}$, where $\text{lcm}(1, 2, \dots, k) = d$ and α_i is the number of i in selected partition.
174. $|Aut(D_{2n})| = n\phi(n)$ where $n \geq 3$.
175. Number of conjugate class of D_{2n} is equal to
- $$\begin{cases} \frac{n+6}{2} & \text{if } n \text{ is even} \\ \frac{n+3}{2} & \text{if } n \text{ is odd} \end{cases}$$
176. $\mathbb{Z}_n[i] \simeq \mathbb{Z}_n \times \mathbb{Z}_n$.
177. If $4|p-3$, then $\mathbb{Z}_p[i]$ is an integral domain.
178. Sum of two subring need not be subring.
179. $\frac{\mathbb{Z}[i]}{\langle a+ib \rangle} \simeq \mathbb{Z}_{a^2+b^2}$ if $\gcd(a, b) = 1$.
180. $\frac{\mathbb{Z}[i]}{\langle n \rangle} \simeq \mathbb{Z}_n[i] \simeq \frac{\mathbb{Z}[i]}{\langle in \rangle}$.
181. $\frac{\mathbb{Z}[x]}{\langle m \rangle} \simeq \mathbb{Z}_m[x]$.
182. $(GF(p^n), +) \simeq \underbrace{\mathbb{Z}_p \times \mathbb{Z}_p \times \dots \times \mathbb{Z}_p}_{n \times \text{times}}$.
183. If $x = a + ib$ either $a = 0$ or $b = 0$ and $|x| = p$ with $4|p-3$, then x is irreducible element of $\mathbb{Z}[i]$.
184. If $x = a + ib$ and $a \neq 0, b \neq 0$ and $a^2 + b^2 = p$, then x is irreducible in $\mathbb{Z}[i]$.
185. $\mathbb{Z}[\sqrt{-d}]$ is not PID if $d > 2$.
186. $\mathbb{Z}[\sqrt{-d}]$ is ED if $d = 1, 2$.

187. $[\mathbb{Q}(a_1^{1/n_1}, a_2^{1/n_2}, \dots, a_k^{1/n_k}) : \mathbb{Q}] = \text{lcm}(n_1, n_2, \dots, n_k)$ where $a_i \neq a_j \neq 1$.
188. Let A be a group. For every element $a \in A$ there exists a unique group homomorphism $\phi_a : \mathbb{Z} \rightarrow A$ that sends $1 \mapsto a$.
189. Let A be a group. Then there exists a bijection between $\text{Hom}(\mathbb{Z}, A)$ and A . i.e., $|\text{Hom}(\mathbb{Z}, A)| = |A|$.
190. For a group G , the map $\text{Hom}(G, G_1 \times G_2) \rightarrow \text{Hom}(G, G_1) \times \text{Hom}(G, G_2)$ is a bijection.
191. For a group G , the map $\text{Hom}(G_1 \times G_2, G) \rightarrow \text{Hom}(G_1, G) \times \text{Hom}(G_2, G)$ is a bijection.
192. Number of group homomorphism $f : \mathbb{Z}_m \rightarrow \mathbb{Z}_n \times \mathbb{Z}_k$ is $\text{gcd}(m, n) \times \text{gcd}(m, k)$.
193. Number of group homomorphism $f : \mathbb{Z}_n \times \mathbb{Z}_k \rightarrow \mathbb{Z}_m$ is $\text{gcd}(m, n) \times \text{gcd}(m, k)$.
194. If $n|m$, then number of onto homomorphism $f : \mathbb{Z}_m \rightarrow \mathbb{Z}_n$ is $\phi(n)$.
195. Number of isomorphism from $f : \mathbb{Z}_n \rightarrow \mathbb{Z}_n$ is $\phi(n)$.
196. Number of homomorphism from $f : \mathbb{Z} \rightarrow \mathbb{Z}_n$ is n .
197. Number of homomorphism from $f : \mathbb{Z}_n \rightarrow \mathbb{Z}$ is 1.
198. Number of homomorphism $f : S_n \rightarrow \mathbb{Z}_m$, $n \geq 3$ is
- $$\begin{cases} 1 & \text{if } m \text{ is odd} \\ 2 & \text{if } m \text{ is even.} \end{cases}$$
199. Number of associates of $a \in \mathbb{Z}_m$ are $\phi(o(a))$.
200. List of associates $cl(a) = \{x \in \mathbb{Z}_m \mid \frac{m}{(x, m)} = o(a)\}$.
201. Number of nilpotent elements in $\mathbb{Z}_{p^n}$ is p^{n-1} .
202. If $m = p_1^{n_1} p_2^{n_2} \dots p_r^{n_r}$, then number of nilpotent elements in $\mathbb{Z}_m$ is $p_1^{n_1-1} p_2^{n_2-1} \dots p_r^{n_r-1}$.